Evolution, 2026, **0(0)**, 1–12
https://doi.org/10.1093/evolut/qpaf240
Advance access publication 12 December 2025
Methods Article

OXFORD

MoleRate: comparing molecular relative evolutionary rates to detect convergent evolution

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Abstract

In comparative evolutionary genomics, faster or slower evolution of a particular gene, site, or branch in a phylogenetic tree, when compared to the appropriate average, has been interpreted as evidence of conservation, functional importance, or adaptation. With large consortia generating hundreds of genomes, there is an opportunity to interrogate these datasets for evidence of accelerated or reduced evolutionary rates in protein-coding genes associated with the presence or absence of a given phenotype (e.g., marine vs. terrestrial, nocturnal vs. diurnal). Such rate shifts can reflect the molecular basis of convergent phenotypic adaptation when they occur repeatedly across independent lineages. Here, we introduce an explicit phylogenetic rate test, MoleRate, for acceleration or reduction of nucleotide or protein evolutionary rates in focal lineages vs. the rest of the phylogeny. Compared to existing methods, MoleRate offers execution, explicit likelihood-based hypothesis testing, and the ability to detect and filter out potentially aberrant signal from single lineages. We demonstrate MoleRate's performance on simulated and empirical data, and apply it to several mammalian phenotypes. We also highlight its visualization capabilities, which enable exploration and communication of results. These analyses show that MoleRate detects biologically significant enrichments in selective pressure on specific functions related to the given phenotype, and that enrichments in selective pressure related to the given phenotype, absent when random lineages are tested.

Keywords: convergent evolution, molecular evolution, phylogenetic comparative methods

Introduction

In adaptation studies, convergently evolving phenotypes serve as natural replicates for evolutionary change across the tree of life. Phenotypic convergence occurs when different species or populations independently evolve similar traits, often due to shared selective pressures (Stern, 2013). At the molecular level, phenotypic convergence can arise through identical amino acid changes, different substitutions within the same gene, or distinct genes providing similar adaptive functions. While convergence at a single amino acid is assumed to be rare (Zhang & Kumar, 1997), gene-level convergence provides increased statistical power for detecting adaptive evolution. One powerful way to detect such patterns is by examining lineage-specific shifts in molecular evolutionary rates, which can signal changes in selective constraint or adaptive evolution.

Methods for detecting convergence focus either on identifying directional selection at specific amino acid sites (Parker et al., 2013; Parto & Lartillot, 2018; Rey et al., 2018) or detecting selection acting on the same gene across species, even if different sites are involved (Martin et al., 2021; Wirthlin et al., 2024). Codon-based tools, such as HyPhy (Kosakovsky Pond et al., 2020; Wisotsky et al., 2020) and codeml (Yang, 2007), are widely used to detect gene-level positive selection and convergent evolution (Barkdull & Moreau, 2023; Shultz & Sackton, 2019; Zehr et al., 2023). However, genome-wide applications of these approaches require substantial computational resources, often necessitating parallelized analyses on computing clusters. Evolutionary rate-based methods, such as RERconverge, PhyloAcc, TRACCER, and TraitRate (Hu et al., 2019; Kowalczyk et al., 2019; Mayrose & Otto, 2011; Treaster et al., 2021), offer a less computationally demanding alternative by detecting shifts in molecular rates across genes. Relative rate approaches have been used since before the beginning of the modern genomic era (Muse & Gaut, 1997), and are computationally relatively cheap (Kowalczyk et al., 2019). Such methods assume that adaptive evolution is typically followed by purifying selection, resulting in lineage-specific rate shifts (Chikina et al., 2016).

While evolutionary rate-based methods are a computationally efficient alternative to codon models, they rely on assumptions that can be improved upon. In particular, RERconverge, one of the most widely used approaches, is best suited for large genome-wide datasets and performs rate

Received April 1, 2025; revisions received October 17, 2025; accepted November 7, 2025
Associate Editor: Tomomi Parins-Fukuchi; Handling Editor: Hélène Morlon
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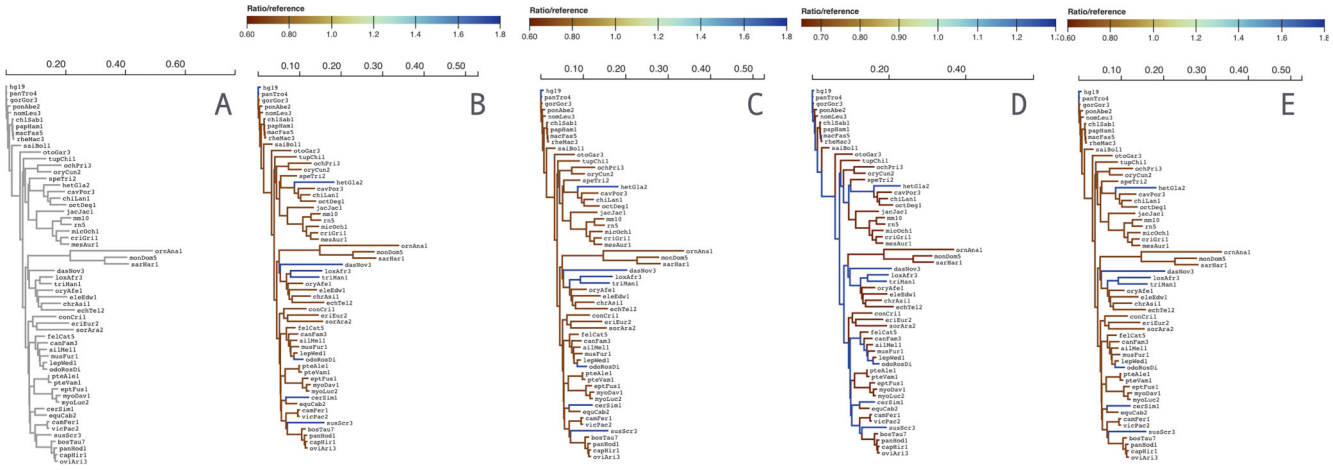

Figure 1. (A) The reference tree (**T_0**) for bird species from Osipova et al. (2024). (B–E) The trees for the *KRT10* gene with branch lengths estimated using the WAG substitution model with different labeling strategies (blue branches). The coloring represents the relative lengths of branches under the *Proportional Partitioned* model (see text) relative to the corresponding length in **T_0**. (B, strategy: tips-only) Only the eight terminal branches representing hairless species are labeled. (C, strategy: all-descendants) In addition to the 8 terminal branches, one internal node is labeled. (D, strategy: some-descendants) In addition to the eight terminal branches, 30 internal nodes are labeled. (E, strategy: parsimony) In addition to the eight terminal branches, one internal node is labeled; in this case the labeling is equivalent to all-descendants.

testing in a two-step framework: estimating rates and then testing them post hoc. This separation can limit power and fails to account for parameter uncertainty.

Here, we introduce MoleRate, a likelihood model-based method designed to detect shifts in evolutionary rates on subsets of phylogenetic tree branches compared to a genome average tree. Implemented in HyPhy, MoleRate is similar in concept to RERconverge, with the key difference that RERconverge performs "standard" regression-type testing on branch length estimates, whereas MoleRate performs "phylogenetic" regression directly as a part of the fitting process. MoleRate tests accommodate several models of rate variation, and implements a procedure to filter out cases when rate variation is attributable to too few outlier branches. We demonstrate the method on several previously studied cases of convergent evolution and simulated data. MoleRate includes intuitive, adaptable JavaScript-based visualization notebooks for interactive result interpretation. Users can efficiently explore rate shifts across genes and lineages and customize the framework for specific datasets and research questions.

Methods

Given a protein or a nucleotide sequence alignment, MoleRate estimates relative evolutionary rates on a designated (focal) part of a phylogenetic tree and performs a series of nested hypothesis tests to determine if rates vary between the focal part of the tree and the rest of the tree, or within the focal branches. If an overall test for rate differences is significant, additional tests can be performed at the level of individual branches to screen for cases where only a single (or a few) branches contribute to the overall difference signal, as an outlier detection test.

Inputs (see Figure 1 for examples)

1. A precomputed (unrooted) reference tree $\mathbf{T}_0$, with branch lengths; the absolute magnitude of the length is not important, as only relative values are relevant. In many applications this tree will be a genome-wide estimate based on a species tree with branch lengths measured in expected substitutions/site.
2. A list of $\mathbf{L}$ (≥ 1) "labeled" species/terminal branches $B = b_1, \ldots, b_L$ present in $\mathbf{T}_0$.
3. A multiple sequence alignment composed of amino acid sequences containing some (but not necessarily all) species/taxa present in $\mathbf{T}_0$. Sequences present in the alignment but not in $\mathbf{T}_0$ are removed.
4. Internal-branch labeling strategy, which can automatically extend B to include additional internal branches. The currently available modes are
   a. labeled-tips: only explicitly specified terminal branches are included
   b. all-descendants: include an internal branch only if all of its descendants are included
   c. some-descendants: include an internal branch if some of its descendants are included
   d. parsimony: decide if an internal branch is included based on two-character parsimony using terminal branch labeling (included = 1, excluded = 0)
5. A substitution model. For protein data, *LG*, *WAG* [default], *JTT*, *JC69*, *mtMet*, *mtVer*, *mtInv*, *gcpREV*, *HIVBm*, *HIVWm*, and the general time reversible (GTR) model are supported. For nucleotide data, MoleRate only implements the GTR model, inspired by the recent work de-emphasizing the custom of nucleotide model selection (Abadi et al., 2019). More generally, the choice of substitution model does not impact relative rate estimation dramatically (Spielman & Kosakovsky Pond, 2018), also see simulations below. All protein models use their +F versions (empirical protein residue frequencies rather than the frequencies derived from training data).
6. Site-to-site rate variation options. Distributions supported: none [default], discretized unit mean gamma, general discrete (GDD). For gamma and GDD, the number of rate classes (2–10, default 4) can be further specified.

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Table 1. The different models available in MoleRate.

Model	Gene branch length expression	Estimated model parameters
Proportional	$\mathbf{g_i = Rt_i}$	**R**, global scaler
Proportional Partitioned	$\mathbf{g_i = R_L t_i}$ if i is labeled	$\mathbf{R_L}$, labeled (focal) scaler
	$\mathbf{g_i = R_0 t_i}$ otherwise	$\mathbf{R_0}$, background scaler
Unconstrained Test	$\mathbf{g_i}$ is unconstrained if i is labeled	$\mathbf{R_0}$, background scaler
	$\mathbf{g_i = R_0 t_i}$ otherwise	**L** independent lengths for each labeled branch
Unconstrained	$\mathbf{g_i}$ is unconstrained	***N*** independent branch lengths
Proportional + 1∗	$\mathbf{g_i = Rt_i}$, if $\mathbf{i \neq k}$, where $\boldsymbol{k}$ is specified	**R**, global scaler
	$\mathbf{g_k}$ is unconstrained	The branch length for branch $\boldsymbol{k}$

Note. (∗) is used for outlier detection.**Proportional.** This model scales all branch lengths from the reference tree by a single estimated factor, which can be interpreted as the relative gene rate (compared to the reference tree; Figure 2A).
Proportional Partitioned. This model scales focal/labeled branch lengths by one factor, and the rest of the branches by an independent factor. In other words, labeled and background branch sets have different "mean" (over branches) relative rates (Figure 2B).
Unconstrained Test. This model scales unlabeled branches by a single estimated factor, and independently estimates all branch lengths in the labeled set. It accommodates relative rate heterogeneity within the labeled set, while assuming homogeneity in the rest of the tree (Figure 2C).
Unconstrained. This is the standard model where all branch lengths are estimated (Figure 2D). The user controls if this model is fitted.
Proportional + 1∗. This model is used for outlier detection (see below). It scales all branches in the tree by a single estimated factor, except for one specified branch in the test set, whose length is estimated independently.

Table 2. Hypothesis testing in MoleRate.

Null model	Alternative model	*d.f.*	Interpretation
Proportional	Proportional Partitioned	1	Evolution is accelerated or slowed on *B* compared to the background. *Gene-wide rate shift on test branches.*
Proportional	Unconstrained Test	*L*	Evolutionary rates are different from the overall gene relative rate on *B*. *Rate heterogeneity across test branches.*
Proportional	Unconstrained∗	*N*	Branch lengths are not proportional to reference. *General rate heterogeneity across the tree.*
Proportional Partitioned	Unconstrained Test	*L*-1	Branch lengths on *B* are not proportional to reference. *Rate heterogeneity within test branches.*
Proportional Partitioned	Unconstrained∗	*N*-2	Group relative rates are constant within the groups. *Rate heterogeneity within test or background branches (or both).*
Unconstrained Test	Unconstrained∗	*N*-*L*-1	Branch lengths are not proportional to reference on background. *Rate heterogeneity within background branches.*

Note. L = number of labeled branches; N = total number of branches; (∗) when fitten.

Fitted models

MoleRate fits branch lengths (and substitution rates for the GTR model) using sequence data from the multiple sequence alignment (MSA) using maximum likelihood. Let $\mathbf{T_0}$, contain N branches, each with the associated numerical branch length $\mathbf{t_i}$, $1 \leq i \leq N$. Four models are available, which differ in the number of parameters and complexity; they differ in how gene-level branch lengths $\mathbf{g_i}$ are estimated (Table 1).

Hypothesis testing and interpretation

MoleRate conducts pairwise likelihood ratio (LR) hypothesis tests between nested models (Table 2). A Holm–Bonferroni multiple testing correction is applied to raw test p-values to control the Type I family-wise error rate.

The key interpretive hypothesis is Proportional (null) vs. Proportional Partitioned (alternative). Given two branch groups, labeled and background, the test determines whether the group level rates (slopes in Figure 2B) are different or not. In evolutionary terms, this test determines if the labeled group evolves at a rate that is slower or faster than the background group. For example, if the branches in the labeled group have a unique phenotype, accelerated protein evolution could indicate positive selection (or relaxation of constraint), and slowed evolution could indicate greater conservation.

Outlier detection

When testing Proportional (null) vs. Proportional Partitioned (alternative), a significant test result that the relative evolutionary rates between labeled and background branches are different may arise due to the influence of a single or a few branches; much like an outlier effect in linear regression (see an example in *Results*). MoleRate implements a simple branch-by-branch testing procedure for such outlier detection. For each branch from the labeled set, $b_k \subset B$, we fit the *Proportional + 1* model (Table 1). Two LR tests (each with a single degree of freedom) are then performed; p-values are corrected using the Holm–Bonferroni procedure.

Single branch. *Proportional* (null) vs. *Proportional + 1* (alternative). A significant result from this test implies that the relative rate on the single branch b_k is different from the background relative rate of the tree. This identifies the branch as having a significant individual rate shift.

All but this branch. *Proportional + 1* (null) vs. *Unrestricted Test* (alternative). A significant result from this test implies that labeled branches in B, except b_k, have sufficiently different relative compared to the background relative rate. This tests for a signal of convergence across the labeled set even after the potential outlier status of b_k is accounted for.

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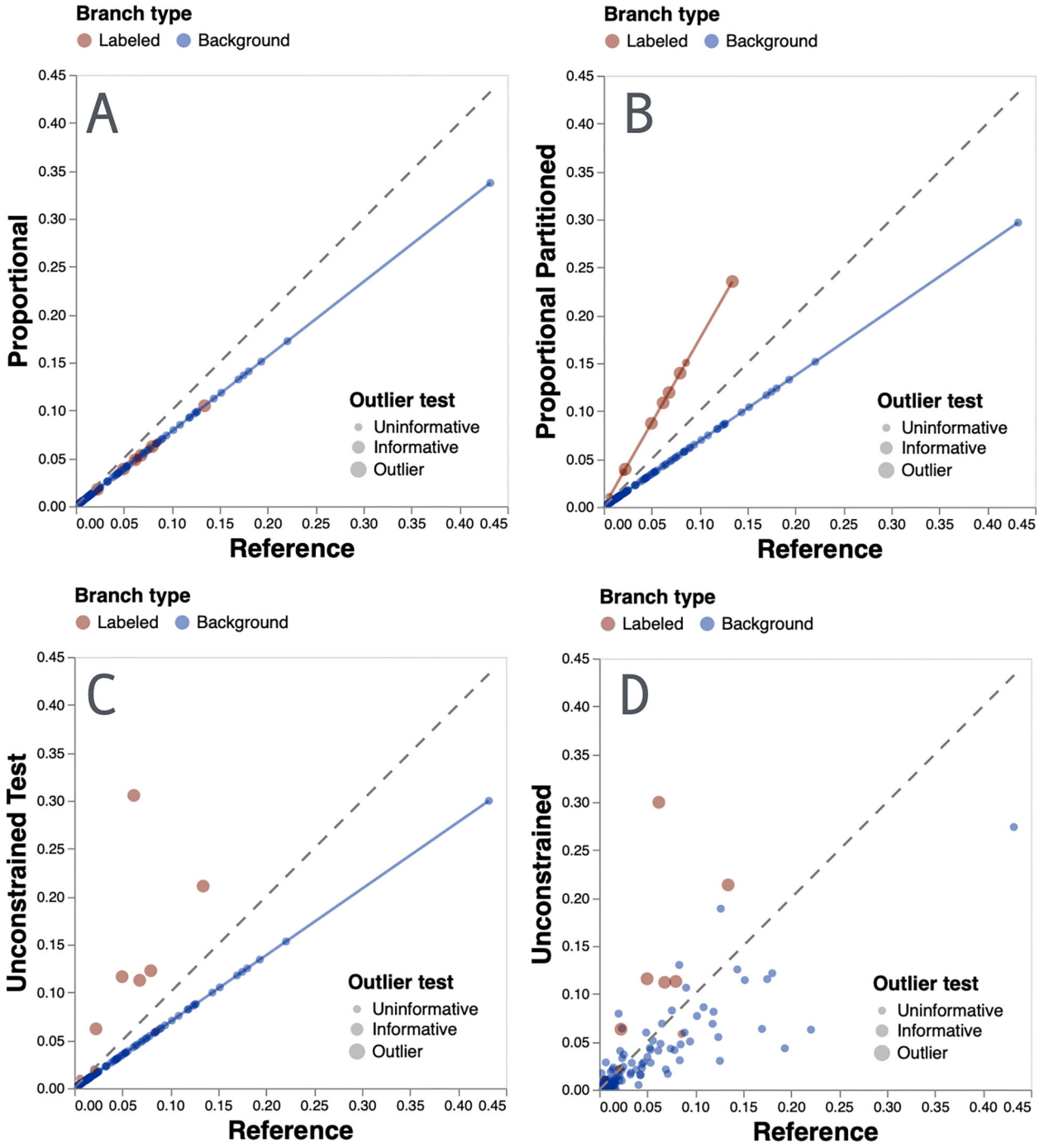

Figure 2. Branch length estimates for the *KRT10* gene (all-descendants labeling strategy) under the four different models compared to the lengths in the reference tree $\mathbf{T_0}$. When estimated branch lengths are constrained to be proportional to the corresponding reference lengths, a line is drawn through the corresponding points for emphasis. For outlier test labels, see *Outlier detection* further in the text.

We classify individual branches as follows:

- **Uninformative.** Neither test is significant, indicating that b_k does not have a significant individual rate shift, nor does its inclusion/exclusion significantly impact the overall convergence result.
- **Informative.** Both tests are significant, indicating that $b_k$and the remaining labeled branches differ from the background.
- **Outlier.** Only the *Single Branch* test is significant, suggesting that $b_k$alone drives the signal of rate difference.

Implementation

MoleRate is implemented in HyPhy versions 2.5.71 or later (Kosakovsky Pond et al., 2020). MoleRate is executed using the hyphy molerate command. MoleRate requires a phylogenetic tree, an input gene alignment, and phenotype annotations to specify foreground and background branches. More details for specific analysis options can be found in the HyPhy Analyses GitHub repository (https://github.com/veg/hyphy-analyses/tree/master/molerate). Alignments are analyzed one at a time, and each analysis produces a machine-readable JSON file, documented in the linked repository.

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Table 3. MoleRate applied to the *KRT10* gene with the subterranean phenotype labeling and different model configurations. RR: relative rate. Mean ranges are shown for models which estimate multiple values. LR p-value: likelihood ratio test p-values, following Holm-Bonferroni correction.

Model	Log-likelihood (ln L)	RR labeled	RR background	LR *p*-value (vs. the corresponding null)		
				Prop.	*Prop. Partitioned*	*Unconst. test*
Proportional	−10,473.6	0.78	0.78	–	–	–
Proportional Partitioned	−10,471.9	1.75	0.69	∼0	–	–
Unconstrained Test	−10,471.1	1.9 [0.7–4.9]	0.69	∼0	∼0	–
Unconstrained	−10,190.2	1.9 [0.7–4.9]	0.69 [0–7.9]	∼0	**∼0**	∼0

Note. RR = relative rate. Mean rates and ranges shown for models, which estimate multiple values. LR *p*-value = likelihood ratio test *p*-values, following Holm–Bonferroni correction.

Each analysis is configured with the following options, also documented in the repository:

1. A protein or nucleotide alignment to analyze.
2. A species tree with branch lengths to be used as reference. MoleRate will compute the intersection of species between the tree and the alignment (case insensitive) and reduce both accordingly. If internal branches are deleted from the tree, their lengths are added to the adjacent branch to maintain relative levels of divergence.
3. A list of focal species. At least one of these species must be in the intersection computed in step 2.
4. Internal branch labeling strategy.
5. Model options: substitution matrix, site-to-site rate variation.
6. Include the Unrestricted model? Yes/no
7. Perform branch level outlier/informative testing? Yes/no

Individual JSON files may be viewed and explored using the following ObservableHQ notebook: https://observablehq.com/@hyphy/molerate. A Python 3 script included in the repository can be used to process directories of MoleRate JSON files and output summary CSV files for subsequent processing or visualization using https://observablehq.com/@hyphy/molerate-summary.

Results

An example: *KRT10* gene in hairless mammals. MoleRate was run with the all-descendants labeling strategy and the default WAG substitution model to obtain the following results (Table 3 and Figure 2). There is a statistically significant acceleration, ∼1.6×, of protein evolutionary rates in the labeled (hairless mammals and their ancestors, Figure 1C). There is also evidence of relative rate heterogeneity within the hairless mammals test set, and across the tree as a whole. Branch-by-branch analysis indicates that six branches (Figure 2C) are the primary drivers of the rate heterogeneity signal; these six branches correspond to African elephant, manatee (and their ancestor), northern white rhinoceros, nine-banded armadillo, and pig. This level of branch-specific insight, provided by the Unconstrained Test model comparisons, is a key advantage of MoleRate over existing methods.

Simulations

We evaluated MoleRate on simulated data, using empirical trees for 40 bird species (Osipova et al., 2024), using the WAG matrix and the *Proportional Partitioned* model. We used the sugar-consuming birds as the labeled set for the avian tree; the labeled set was either just the terminal branches (*tips*), or the terminal branches plus the ancestral branches (*tips + anc*) labeled with the all-descendants strategy (Figure S1). We set the background branch relative rate to 1, and varied the relative rate for the labeled branches from 0.1 to 3.0 in increments of 0.1. The length of the alignments was varied from 100 to 1,000 in increments of 100. For each combination of the relative rate and alignment length, we generated 100 replicates, yielding a total of 30,000 simulated alignments per simulation scenario. Simulated alignments are available from https://github.com/veg/molerate.

For the primary MoleRate test (Proportional:Proportional Partitioned models, Figure 3), we observe that

1. Under the null scenario (no difference in rates between branches), all combinations of lengths, relative rates, and labeling strategies yielded nominal or conservative performance (detection rate, or Type I error of ≤ 0.05 for $p = 0.05$).
2. Power of the test (detection rate) increased both with alignment length (∼sample size) and as the relative evolutionary rate moved further away from 1 (∼effect size). By the time the relative rate was outside [0.8, 1.5], all but the shortest alignments had power >80%, and with rates outside [0.5, 2.0] effectively all scenarios had power ∼100%.
3. When the branch labeling scenario was incorrectly specified, there was a small loss of efficiency (power) compared to the correctly specified labeling scenario. When tips-only was selected for tips + ancestors simulations, this is mostly due to having fewer branches to test. When the all-descendants option was specified for tips simulations, this is because ancestral branches added to the test set were evolving at background rates.
4. The (Proportional:Unrestricted test models, Figure S1), similar observations hold, except that the test has less power because the alternative model is overparameterized.
5. The (Proportional Partitioned:Unrestricted test models, Figure S2) evaluates whether branches in the labeled set evolve at rates that differ proportionally from the background. When the correct model is fitted (Figure S2A and D), the test appropriately returns low detection rates. When an incorrect model is fitted, particularly under the "all descendants" labeling strategy, the Unrestricted test model is preferred to the Proportional Partitioned model, as it can better accommo-

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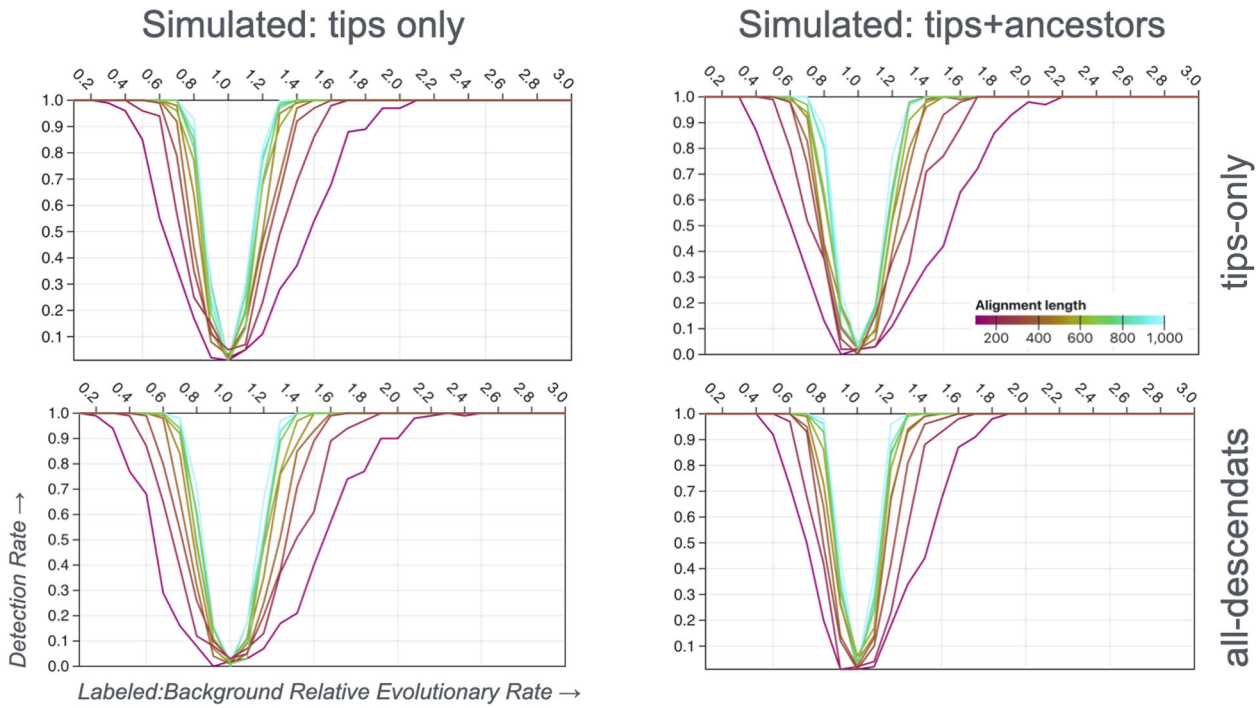

Figure 3. Detection rate (fraction of 100 replicates with p-value ≤ 0.05 for a given scenario) for the Proportional:Proportional Partitioned test, as a function of simulated alignment length (color), the labeled/background branch rate ratio (**R_L varied, R_0** $= 1$, x-axis), and the labeling strategy (all-descendants, or tips-only; table row). We considered simulations where the labeled set was just the terminal branches (tips only, Figure 1B) or terminal branches and their ancestors (Figure 1C). The WAG evolutionary model was used to simulate and infer.

date the rate variation introduced by the mislabeled branches (Figure S2C).

6. Using a misspecified substitution model (JC69 instead of WAG) has minimal impact on model performance (Figure S3), consistent with previous results in similar contexts (e.g., relative rates for individual alignments sites; Spielman & Kosakovsky Pond, 2018).

Empirical protein dataset analysis

We applied MoleRate to a collection of 19,149 gene alignments on 62 mammalian species (not every gene had every species), based on the University of California, Santa Cruz (UCSC) genome alignments (Blanchette et al., 2004; Harris, 2007; Kent et al., 2002). A version of this dataset was previously analyzed using RERConverge (Kowalczyk et al., 2019) and other tools with three different discrete phenotypes: marine (Chikina et al., 2016), subterranean (Partha et al., 2017), and hairlessness (Kowalczyk et al., 2022). As a control, we also selected a set of six random species to label. The original codon-level alignments were translated into corresponding protein sequences. The results of the MoleRate analysis on these phenotypes are summarized in Table 4.

We labeled the appropriate species (Figure S4) and ran the relative rate analysis with the WAG substitution model and site-to-site rate variation modeled as a three-bin general discrete distribution. We used the none internal branch labeling strategy. For each discrete phenotype, we identified candidate sets of accelerated and decelerated genes, using the following criteria:

1. Benjamini–Hochberg false discovery rate (FDR) (Benjamini & Hochberg, 1995) for the Proportional:Proportional Partitioned test is ≤ 0.01.
2. If N total branches are labeled, then the number of tested and individually informative labeled branches (see below) is at least $N/2$.
3. Zero outlier branches are detected (at $p \leq 0.01$), ensuring that the observed signal is not driven by a single lineage but is instead shared across multiple labeled branches.
4. Genes meeting conditions 1–3 were ranked by the relative evolutionary rate ($\mathbf{R_L/R_0}$) on the labeled branches versus the rest of the tree (Proportional Partitioned model). *Accelerated* genes had $\mathbf{R_L/R_0 \geq 2}$ and *decelerated* genes had $\mathbf{R_L/R_0 \leq 0.5}$.

This particular set of criteria was selected to provide a limited list of strongest (low FDR, multiple involved lineages, large effect site) candidate genes to investigate. Other selection criteria are easily conceived: any genes passing FDR, allowing some outliers etc.

Outlier detection. To evaluate whether significant rate shifts were driven by single lineages or shared among multiple branches, we applied MoleRate's outlier detection procedure (Figure S5). Among the significantly accelerated genes, *LIM2* (lens intrinsic membrane protein 2) represents a well-supported case: all three tested (subterranean) branches provide strong evidence for rate acceleration. The hypotheses that each branch has the same rate as the background (*One Branch* test) can be rejected, and the hypotheses that the remaining branches (*All but this branch* test) have the same rate as the background can be similarly rejected. In contrast, *SEPT7* (filament-forming cytoskeletal GTPase), a sin-

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Table 4. MoleRate applied to the UCSC gene alignments for four different labeled branch sets: subterranean was run twice—one as protein data and once as nucleotide data.

Phenotype	Labeled branches	Tested	FDR ≤ 0.01	Accelerated	Decelerated	Outliers
Subterranean	3	17,997	2,492	113	131	1,304
Subterranean (nucleotides)	3	17,963	3,743	60	44	1,954
Marine	5	18,855	3,719	120	148	1,514
Hairless	10	19,178	4,808	39	37	1,121
Random	6	19,164	2,520	65	27	983

Note. FDR = false discovery rate. Labeling strategy for internal branches was set to none. Tested: genes with at least two labeled branches present; FDR ≤ 0.01: genes where the Proportional:Proportional Partitioned test returned a q-value of ≤ 0.01; accelerated: genes deemed accelerated based on the three criteria in the text; decelerated: genes deemed decelerated based on the three criteria in the text; outliers: genes with FDR ≤ 0.01, but where the significant result is driven by a single "outlier" branch.

gle branch (star-nosed mole) provides the entire rate difference signal ($\mathbf{R_L/R_0} > 4$); notably there is no evidence that the remaining two branches evolve at rates different from background. A manual inspection of the corresponding gene alignment shows an obvious sequence assembly/alignment error in this species toward the 3′ end of the gene. Masking this misalignment completely removes the rate difference signal (results not shown).

Subterranean phenotype. To further characterize the functional relationships among accelerated and decelerated genes in subterranean lineages, we performed protein–protein interaction network analyses using STRING v12.0 (Szklarczyk et al., 2023). We uploaded the top accelerated (Table S1) and decelerated (Table S2) genes separately and examined enrichment for functional categories, biological processes, and pathways using default parameters (minimum required interaction score = 0.4, medium confidence; evidence from experiments, databases, and text mining included).

The top accelerated genes in the three subterranean branches are significantly enriched for eye formation, vision, and olfaction; most decelerated genes are also enriched for olfactory receptors and sensory perception (Table 5). Loss of vision in subterranean mammals coupled with enhanced olfaction is well-documented in previous studies (Davies et al., 2018; Nevo, 1979; Partha et al., 2017); these and other studies also reported immune, skin, morphological, and hypoxia-related adaptation. Genes involved in lens/eye development and olfaction are predominant at the top of the list for most accelerated genes (Table S1). Acceleration of protein substitution rates would be consistent with two distinct processes: loss of function (e.g., due to loss of vision), leading to relaxation of selection, or gain of function/adaptation, leading to positive selection. Protein evolutionary rates alone cannot differentiate between these two scenarios, and other tests, e.g., based on dN/dS or other adaptive evolution measures are traditionally used for this purpose. Deceleration of protein evolutionary rates in subterranean species could be interpreted as intensified purifying selection, possibly following a burst of adaptation (Table S2).

As an ancillary analysis, we ran codon-based test of selection relaxation/intensification (Wertheim et al., 2015) on the genes listed in Tables S1 and S2. For subterranean accelerated genes, 80/113 genes were experiencing significantly relaxed selection (RELAX $k < 1$) compared to the rest of the lineages at $p \leq 0.05$ (41/113 at $p \leq 0.00001$), suggesting that loss of constraint is a key driver of rate acceleration (Table S1). Only two genes with evidence of accelerated protein rates on subterranean branches (*GSDMB* and *OR5D14*) had statistically significant test results for selection intensification (RELAX $k > 1, p \leq 0.05$).

Conversely, for subterranean decelerated genes, 82/131 genes were experiencing significantly intensified selection (RELAX $k < 1$) compared to the rest of the lineages at $p \leq 0.05$ (7/131 at $p \leq 0.00001$), suggesting that *increased* constraint is a significant driver of slower rates (Table S2). None of these 131 had statistically significant test results for selection relaxation ($p \leq 0.05$).

Marine phenotype. The top accelerated genes in the five marine branches are significantly enriched for olfactory receptor activity and sensory perception (Table 5). The top decelerated genes are related to heterochromatin assembly and regulation, sperm motility and fertilization. Relaxation of constraint on genes related to olfaction is a well-characterized evolutionary process for marine mammals (Chikina et al., 2016; Farina et al., 2024; Kishida & Thewissen, 2012). Gene ontology and pathway enrichment were analyzed using STRING-DB (minimum required interaction score = 0.4, medium confidence; evidence from experiments, databases, and text mining included), following the same procedure as for the marine phenotype.

For marine accelerated genes, 82/120 genes were experiencing significantly relaxed selection (RELAX $k < 1$) compared to the rest of the lineages at $p \leq 0.05$ (45/120 at $p \leq 0.00001$), consistent with loss of constraint being a major factor in rate acceleration (Table S3). Five genes, however, showed evidence for intensified selection pressure ($k > 1$ and $p \leq 0.05$): *NT5DC4*, *ZNF835*, *SBSN*, *ZNF416*, and *OR4S1*.

On the other hand, for marine decelerated genes, 25/148 genes were experiencing significantly intensified selection (RELAX $k > 1$) compared to the rest of the lineages at $p \leq 0.05$ (1/148 at $p \leq 0.00001$; Table S4). Two of the genes showed evidence for relaxed selection pressure ($k < 1$ and $p \leq 0.05$): *SHISA8* and *CNST*.

Hairless phenotype. The top accelerated genes in the 10 hairless branches are significantly enriched for terms and pathways related to epidermis and hair formation, olfactory receptor activity, an infection pathway, and an estrogen signaling pathway (Table 5). The top decelerated genes in the same branches are enriched for the herpes simplex virus 1 infection KEGG pathway. Gene ontology and pathway enrichment were analyzed using STRING-DB (minimum required interaction score = 0.4, medium confidence; evidence from experiments, databases, and text mining included), follow-

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Table 5. STRING-DB (Szklarczyk et al., 2023) enrichment analysis (using *H. sapiens* reference) for the genes GO biological process/molecular function, and KEGG and Reactome pathways applied to the sets of accelerated and decelerated genes for the tested phenotypes (labeling strategy = none).

Phenotype	Gene group	Molecular function	Counts	$\log_{10}$ enrichment	FDR
Subterranean	Accelerated	Structural constituent of eye lens	6/24	1.66	$\sim 10^{-5}$
		Lens development in camera-type eye	9/81	1.47	$\sim 10^{-6}$
		Detection of visible light	5/47	1.29	0.007
		Visual perception	19/223	1.2	$\sim 10^{-13}$
		Odorant binding	8/109	1.13	0.0002
	Decelerated	Expression and translocation of olfactory receptors (Reactome)	33/383	1.1	$\sim 10^{-21}$
		Olfactory receptor activity	31/397	1.09	$\sim 10^{-20}$
		Olfactory transduction (KEGG)	30/417	1.08	$\sim 10^{-22}$
		Sensory perception of chemical stimulus	32/503	1.0	$\sim 10^{-19}$
Subterranean (nucleotide)	Accelerated	Structural constituent of eye lens	4/24	1.77	0.001
		Lens development in camera-type eye	6/81	1.42	0.0002
		Odorant binding	7/109	1.35	$\sim 10^{-5}$
		Olfactory transduction (KEGG)	13/417	1.04	$\sim 10^{-7}$
		Expression and translocation of olfactory receptors (Reactome)	13/383	1.08	$\sim 10^{-7}$
	Decelerated	Expression and translocation of olfactory receptors (Reactome)	12/383	1.21	$\sim 10^{-8}$
		Olfactory receptor activity	12/397	1.2	$\sim 10^{-8}$
		Olfactory transduction (KEGG)	12/417	1.17	$\sim 10^{-8}$
		Detection of stimulus involved in sensory perception	13/515	1.12	$\sim 10^{-7}$
Marine	Accelerated	Expression and translocation of olfactory receptors (Reactome)	30/383	1.13	$\sim 10^{-21}$
		Olfactory receptor activity	30/397	1.11	$\sim 10^{-19}$
		Olfactory transduction (KEGG)	30/417	1.09	$\sim 10^{-21}$
		Detection of chemical stimulus involved in sensory perception	31/445	1.08	$\sim 10^{-19}$
	Decelerated	Sperm motility and taxes (Reactome)	5/9	1.88	0.0001
		Regulation of DNA methylation-dependent heterochromatin assembly	4/19	1.46	0.02
		Regulation of heterochromatin assembly	5/29	1.38	0.02
		Fertilization	6/26	1.5	0.0001
Hairless	Accelerated	Structural constituent of skin epidermis	4/37	1.77	0.002
		Intermediate filament organization	5/69	1.6	0.0007
		Formation of the cornified envelope (Reactome)	7/128	1.48	$\sim 10^{-5}$
		Staph. aureus infection (KEGG)	3/86	1.28	0.05
		Estrogen signaling pathway (KEGG)	4/133	1.22	0.0123
		Expression and translocation of olfactory receptors (Reactome)	9/383	1.11	$\sim 10^{-5}$
		Olfactory receptor activity	9/397	1.09	0.0002
		Olfactory transduction (KEGG)	9/417	1.07	$\sim 10^{-5}$
	Decelerated	Herpes simplex virus 1 infection (KEGG)	10/478	1.06	$\sim 10^{-6}$

Note. Counts: number of genes in the corresponding set with the annotation; number of all genes with the annotation. Only categories with ≥10× enrichment are shown, entries are sorted by decreasing enrichment. FDR: false discovery rate reported by STRING-DB.

ing the same procedure as for the marine and subterranean phenotypes.

For hairless accelerated genes, 30/39 genes were experiencing significantly relaxed selection (RELAX $k < 1$) compared to the rest of the lineages at $p \leq 0.05$ (24/39 at $p \leq 0.00001$), consistent with loss of constraint being a major factor (Table S5). On the other hand, for marine decelerated genes, 20/37 genes were experiencing significantly intensified selection (RELAX $k > 1$) compared to the rest of the lineages at $p \leq 0.05$ (6/37 at $p \leq 0.00001$), consistent with a significant role for constraint for slower rates (Table S6). One gene (*CNST*) in this group showed significant evidence of relaxation ($p \leq 0.05$). Relaxation of constraint on genes and pathways involved with hair formation and development in hairless mammals is an obvious and well-supported result (Kowalczyk et al., 2022; Nery et al., 2014; Sun et al., 2017). Loss of constraint on olfactory genes is most likely a confounder with the marine mammal phenotype (also hairless). Because we required that ≥5 branches independently support acceleration/deceleration, the set of genes meeting selection criteria was much smaller than for the other two phenotypes.

Random subset. While many individual genes met the selection criteria, no functional or pathways categories were sufficiently enriched for an analysis using six randomly chosen subspecies, even though a substantial number of individual genes passed selection criteria (Table 4).

The effect of changing filtering criteria. To assess how different post-analysis filtering choices influence downstream biological interpretation, we examined how the number and composition of genes identified as accelerated depend on the criteria used to define them. Specifically, we compared results

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Table 6. Accelerated (ratio ≥ 2) genes found MoleRategenes as a function of gene selection criteria.

Selection criteria	Subterranean		Marine	
	Genes	Top enriched category	Genes	Top enriched
FDR ≤ 0.01	316	Structural constituent of eye lens (13/24)	838	Expression and translocation of olfactory receptors (Reactome 113/383)
FDR ≤ 0.01 & >1 informative branch	114	Visual perception (19/233)	302	Expression and translocation of olfactory receptors (Reactome 66/383)
FDR ≤ 0.01 & 0 outliers	124	Visual perception (20/233)	325	Expression and translocation of olfactory receptors (Reactome 68/383)
FDR ≤ 0.01 & 0 outliers & >1 informative branch	113	Visual perception (19/233)	291	Olfactory signaling pathway (Reactome 68/417)

Note. The "top" enriched category (significant FDR and largest STRING-DB, Szklarczyk et al., 2023, reported signal [weighted harmonic mean between the observed/expected ratio and -log(FDR)]), and the number of accelerated genes/total category genes are shown.

obtained under progressively stricter definitions of convergence, requiring multiple informative branches and excluding outlier-driven cases. As a simple sensitivity analysis, we considered the size and composition of accelerated gene sets for the subterranean and marine phenotypes as a function of different selective criteria (Table 6). Encouragingly, enrichment analyses are semantically congruent. Requiring more than one informative branch is similar, but not identical to requiring 0 outlier branches. More than 50% of genes with significantly accelerated rates are fully explained by the influence of a single branch (compare FDR ≤ 0.01 to other categories). Existing literature and the overall logic of convergent evolution analyses emphasizes events shared or repeated across different lineages/branches. How to balance enlarging the set of candidate genes (recall), and more targeted selection of candidate genes (precision) is a decision that needs to be made individually, study by study.

Perhaps the strongest argument against not requiring multiple informative branches (and 0 outliers) is the influence of sequencing and other types of errors (see the example above).

Nucleotide vs. amino acid rates. For comparison, we ran MoleRate on the subterranean phenotype but treating the alignments as nucleotide data. Because of lower levels of conservation in nucleotide space and the large influence of synonymous substitutions on branch lengths, it is not clear a priori what level of agreement one would expect between the two rate analyses. While the raw number of significantly accelerated/decelerated genes found by a nucleotide analysis is higher (Table 4), filtering by multiple branches and outliers yields much smaller (∼50%) gene sets compared to a protein analysis. Specifically, of the top 60 accelerated genes identified by the nucleotide analysis, 44 genes were also top accelerated genes in the amino acid subterranean test. Similarly, of the top 44 decelerated genes from the nucleotide analysis, 26 genes were top decelerated genes in the amino acid subterranean test. The smaller nucleotide gene sets are enriched for essentially the same categories as the corresponding protein gene sets.

Computation performance. Because MoleRate fits relatively simple models, and in many cases does not even need to estimate individual branch lengths, meaning its run times are modest. For example, the UCSC subterranean phenotype analysis required a median of 12 s/alignment, including outlier tests on an Apple M4 processor (two threads). Given the embarrassing parallelism of the analysis, the entire genomic analysis would require <1 hr on a modern compute node with 256 total cores, for instance.

Comparison to other similar methods

We compared the performance of MoleRate to RERconverge. Because RERconverge depends on background branches to estimate the mean and variance of evolutionary rates across the tree, we hypothesized that its performance may be reduced when the number of background species is small. Since MoleRate models branch-specific rates directly within a likelihood framework, we further hypothesized that it may be less sensitive to background size and could perform better under these conditions. To test these hypotheses, we analyzed the subterranean and hairless trait using both methods, comparing results on the full UCSC species set and on pruned species trees. Figure 4 shows precision–recall curves comparing the performance of RERConverge and MoleRate using the subterranean trait and a tree with 20 background species. These enrichment tests include four candidate gene sets, including: Lens Morphology Genes, Retina Morphology Genes, Eye Physiology Genes, and Expressed Ocular Genes. The phenotype-based sets were obtained from the MGI Mammalian Phenotype database via the Enrichr interface (cataract [MP:0001304] and abnormal lens morphology [MP:0001303] for lens; abnormal retina morphology [MP:0001325] for retina; abnormal eye physiology [MP:0005253], abnormal cone electrophysiology [MP:0004022], abnormal eye electrophysiology, and abnormal rod electrophysiology for physiology). The expressed ocular gene sets were those with protein expression in the human eye confirmed by liquid chromatography-tandem mass spectrometry (LC-MS/MS) (Ahmad et al., 2018) and no expression outside ocular tissues in Human Protein Atlas (Karlsson et al., 2021; Yu et al., 2015), resulting in 65 eye-specific genes.

With only 20 background species, MoleRate recovered greater enrichment of candidate genes at similar levels of recall compared to RERconverge, particularly for the lens morphology and expressed ocular gene sets (Figure 4). This advantage was even more pronounced when using MoleRate's outlier testing, which ensures that signals are not driven by a single lineage but instead reflect convergence across multiple labeled branches. By contrast, when analyses were performed using the full species set, both methods showed more comparable performance (Figure S6).

When comparing the performance of two methods, interpretation is challenging because the "true" set of convergently evolving genes is not known in advance. In the subterranean case, it might seem encouraging if one method detects stronger signal among ocular genes, but more signal does not necessarily imply greater accuracy. What we can say

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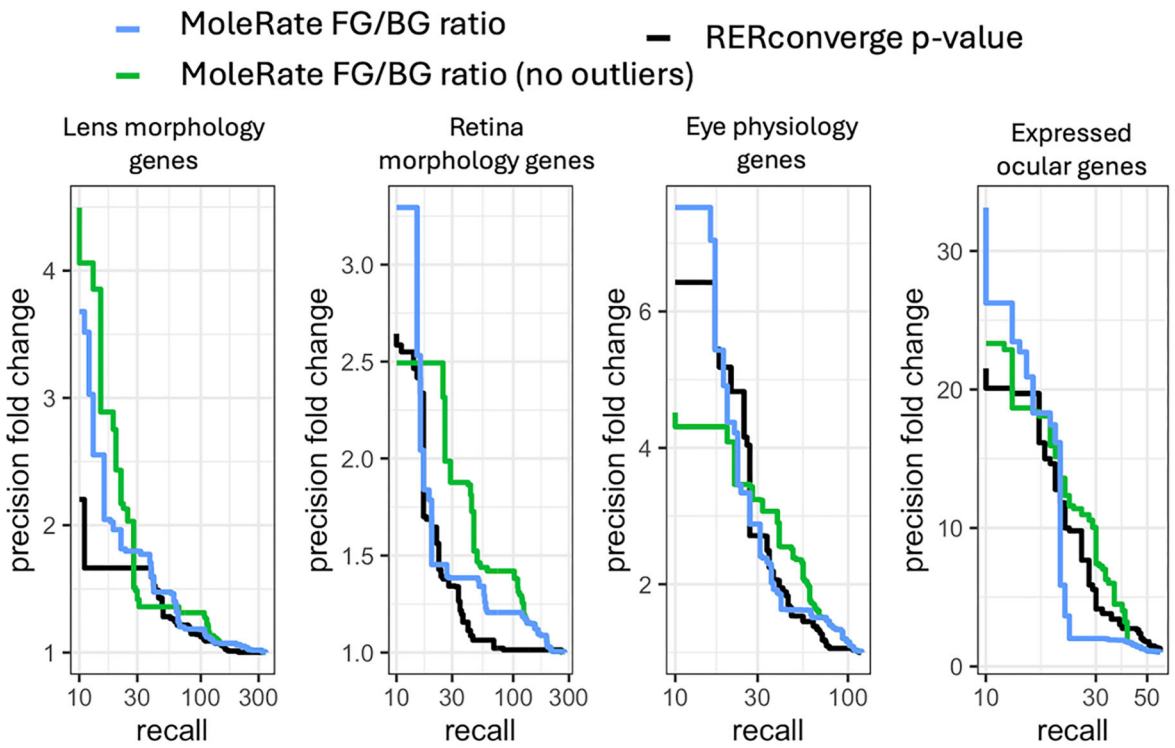

Figure 4. Precision–recall curves comparing the performance of RERConverge (black curve) and MoleRate (blue, green curves, representing results after FDR correction) on the subterranean phenotype using all background species, across four candidate gene sets (Lens Morphology, Retina Morphology, Eye Physiology, and Expressed Ocular). The x-axis shows recall, defined as the proportion of genes in each candidate set recovered as significant by the evolutionary analysis. The y-axis shows precision fold-change, calculated as the log-transformed enrichment of candidate genes among the significant results relative to expectation.

with confidence is that both RERConverge and MoleRate produce similar overall performance curves, and both detect enough vision-related genes for enrichment analyses to consistently recover eye-related functions, with MoleRate being more precise when a small set of high-confidence genes is sought.

Discussion

We developed and benchmarked a phylogenetic tool, MoleRate, to identify genes or other alignable elements in multispecies genome-scale alignments, where nucleotide or amino acid (coding data only) substitution rates are significantly different between pre-designated sets of branches in the species tree. Using a reference tree with genome-averaged branch lengths makes this analysis fast and applicable to rapid screening of genomic datasets with modern scales. Finding genes where specific lineages/branches are evolving much faster or much slower than the rest of the tree is relevant for many biological applications, the key one being: which genes are associated with the development or a specific phenotype or trait in multiple independent branches or lineages?

Comparison to existing relative rates methods. There are several existing tools and approaches for performing these types of analyses, either based on rate estimation, dN/dS analysis, or more complex phylogenetic regression. MoleRate approach is not conceptually new, but it offers several unique features that motivate its development. Compared to RERConverge, which takes estimated rates and performs *post hoc* analyses on them, MoleRate performs rate estimation and testing within the phylogenetic inference framework, thereby automatically accounting for parameter estimation uncertainty. Compared to dN/dS and Bayesian methods used in this context, MoleRate is several orders of magnitude faster (seconds per gene), and less sensitive to errors in alignment and model misspecification. MoleRate includes tests for outlier signal, so that reported rate acceleration or deceleration is robustly supported by multiple tree branches. It also directly estimates the effect size, i.e., the extent of acceleration or deceleration, and allows for more fine-grained testing of within-group variation in rates. In addition, MoleRate is accompanied by extensive visualization tools that enable exploration of both individual gene analysis results and large-scale genome-wide analysis results. Our visualization tools extend well beyond what is typically offered by other similar tools.

Performance on simulated and empirical data. MoleRate performs well on simulated data, with power to detect acceleration and deceleration growing with effect size and alignment size, and this performance is minimally affected by substitution model misspecification, or some errors in how branches are labeled. Using an empirical benchmark dataset, a UCSC alignment of 62 mammal species, MoleRate reliably recovers many accelerated and decelerated genes found by previous studies using different approaches. It recovers very clean signals for enrichment of expected functional and molecular categories: vision (loss) and olfaction in subterranean species, olfaction (loss) in marine species, and hair (loss) in hairless species. Acceleration and deceler-

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ation in protein space is largely attributable to relaxation or intensification of selective constraint, validated here with a codon-based RELAX test. Top enriched categories are robust to changes in how genes are filtered, and even the use of nucleotide data for testing. An important observation following from MoleRate analyses is that as much as 50% of the statistically significant signals for rate changes are driven by a single branch (species), which could be indicative of alignment, assembly, or annotation errors or other aberrations.

The simplicity and speed of MoleRate coupled with its robust recovery of expected genes and features from extensively analyzed phenotypes position the tool as a useful addition for screening and evolutionary annotation of large genomic datasets. Production of candidate genes and pathways for further analyses and annotation in the context of phenotypic or convergent evolution is a useful hypothesis generation tool.

Supplementary material

Supplementary material is available online at *Evolution*.

Data availability

All data and code associated with this study are publicly available. Current versions of the datasets used in our analyses can be accessed at the UCSC genome browser https://genome.ucsc.edu/ and the specific alignments used by us here, along with the results can be found at https://github.com/veg/molerate. MoleRate is available as part of the HyPhy package (≥2.5.71), which can be downloaded at https://github.com/veg/hyphy along with additional scripts for processing. Notebooks for data visualization are available at https://observablehq.com/@hyphy/molerate. Data and code will remain accessible to ensure reproducibility.

Author contributions

A.G.S. conducted method testing and contributed to manuscript writing. N.C. conceived the original idea for the project, edited the manuscript, and verified the biological interpretation of results. A.N. contributed to manuscript editing. M.C. conceived the original idea for the project, performed statistical analyses to confirm results, and edited the manuscript. S.L.K.P. wrote portions of the manuscript, and developed the software. All authors reviewed and approved the final manuscript.

Funding

This work was supported in part by the following awards: NIH/NHGRI (HG009299), NIH/NIGMS534 (GM151683), and NIH/NIAID (AI183870).

Conflict of interest

The authors declare no conflicts of interest related to this work. Editorial processing of the manuscript was conducted independently of N.C., who is an Associate Editor of *Evolution*.

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Received April 1, 2025; revisions received October 17, 2025; accepted November 7, 2025
Associate Editor: Tomomi Parins-Fukuchi; Handling Editor: Hélène Morlon
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